Illustration 10.1

Let us make an attempt to find the slope of the tangent line to the graph of $f(x) = x^2$ at (1, 1).

As a start, let us take $\Delta x = 0.1$ and find the slope of the secant line through $(1,1)$ and $(1.1,(1.1)^2)$.

(i) $f(1.1) = (1.1)^2 = 1.21$

(ii) $\Delta y = f(1.1) - f(1)$

$= 1.21 - 1 = 0.21$

(iii) $\dfrac{\Delta y}{\Delta x} = \dfrac{0.21}{0.1} = 2.1$

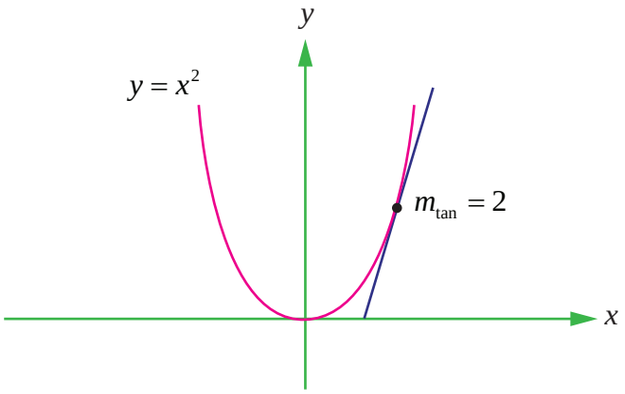


Fig. 10.14

Tabulate the successive values to the right and left of 1 as follows :

Δx	$1 + \Delta x$	$f(1)$	$f(1 + \Delta x)$	Δy	$\Delta y / \Delta x$
0.1	1.1	1	1.21	0.21	2.1
0.01	1.01	1	1.0201	1.0201	2.01
0.001	1.001	1	1.002001	0.002001	2.001
– 0.1	0.9	1	0.81	– 0.19	1.9
– 0.01	0.99	1	0.9801	– 0.0199	1.99
– 0.001	0.999	1	0.998001	– 0.001999	1.999

Clearly, $\lim\limits_{\Delta x \to 0^-} \dfrac{\Delta y}{\Delta x} = 2$; $\lim\limits_{\Delta x \to 0^+} \dfrac{\Delta y}{\Delta x} = 2$.

This shows that $\lim\limits_{\Delta x \to 0} \dfrac{\Delta y}{\Delta x} = 2$.

Thus the slope of the tangent line to the graph of $y = x^2$ at (1, 1) is $m_{tan} = 2$.

On the basis of the Fig.10.6 to 10.13, Illustration 10.1, and our intuition, we are prompted to say that if a graph of a function $y = f(x)$ has a tangent line L at a point P, then L must be the line that is the **limit** of the secants PQ through P and Q as $Q \to P$ ($\Delta x \to 0$). Moreover, the slope m_{tan} of L should be the **limiting** value of the values m_{sec} as $\Delta x \to 0$. This is summarised as follows:

Definition 10.1 (Tangent line with slope *m*)

Let f be defined on an open interval containing x_0, and if the limit

$\lim\limits_{\Delta x \to 0} \dfrac{\Delta y}{\Delta x} = \lim\limits_{\Delta x \to 0} \dfrac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = m_{tan}$ exists, then the line passing through $(x_0, f(x_0))$ with slope m is the **tangent line** to the graph of f at the point $(x_0, f(x_0))$.

The slope of the tangent line at $(x_0, f(x_0))$ is also called the **slope of the curve** at that point.

The definition implies that if a graph admits tangent line at a point $(x_0, f(x_0))$ then it is unique since a point and a slope determine a single line.

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